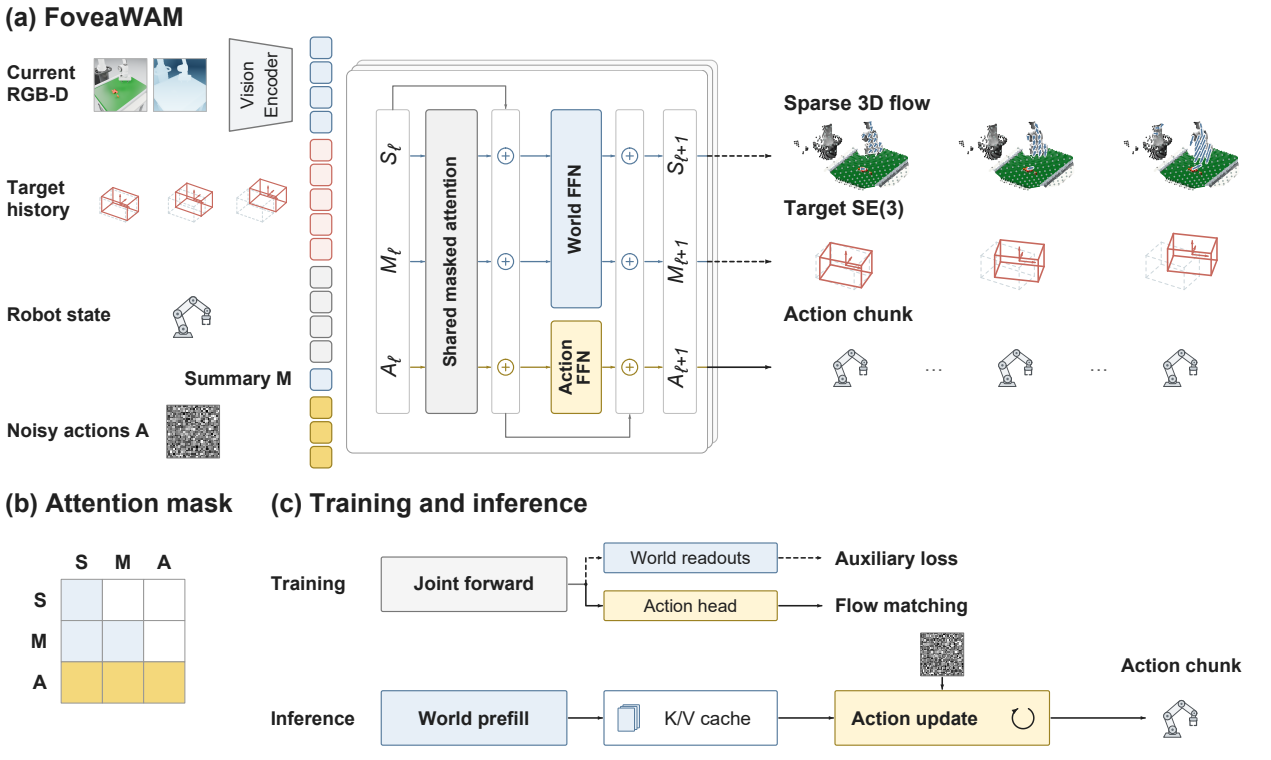
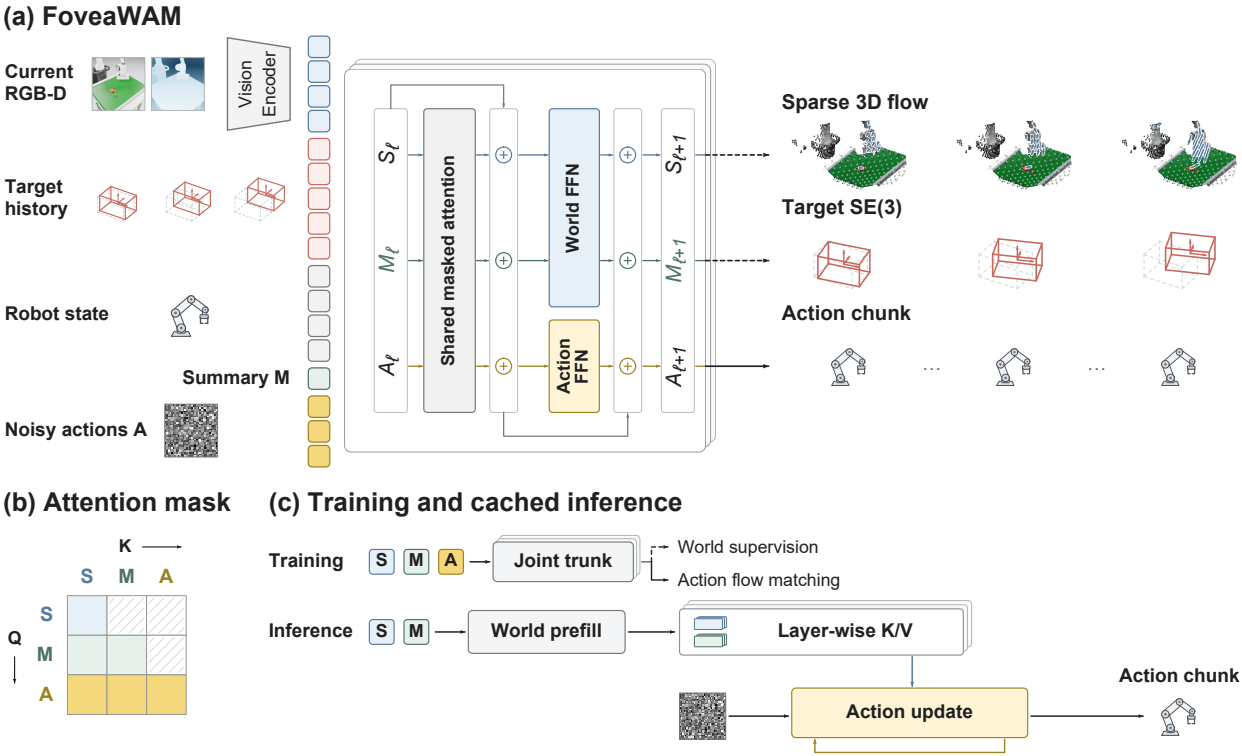

(a) FoveaWAM
Current RGB-D
Vision Encoder
Target history
Robot state
Summary M
Noisy actions A
S_ℓ
M_ℓ
A_ℓ
Shared masked attention
World FFN
Action FFN
$S_{\ell+1}$
$M_{\ell+1}$
$A_{\ell+1}$
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
S
M
A
(c) Training and inference
Training
Joint forward
World readouts
Auxiliary loss
Action head
Flow matching
Inference
World prefill
K/V cache
Action update
Action chunk


(a) FoveaWAM
Current RGB-D
Vision Encoder
Target history
Robot state
Summary M
Noisy actions A
S_ℓ
M_ℓ
A_ℓ
Shared masked attention
World FFN
Action FFN
$S_{\ell+1}$
$M_{\ell+1}$
$A_{\ell+1}$
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
K
S
M
A
Q
S
M
A
(c) Training and cached inference
Training
S
M
A
Joint trunk
World supervision
Action flow matching
Inference
S
M
World prefill
Layer-wise K/V
Action update
Action chunk